**Unit no. ° 15**

# CONSOLIDATION AND TRANSITION TO PREMIÈRE

25 May·11 June 2027

## I Review of calculation, functions and geometry

### Example 15.1: Annual Review · Part A

1. Make 168 /252 in lowest terms; calculate $2^{-3} + 1/4$.

2. Factorise $x^2 - 6x + 9$, then solve $(x - 3)^2 = 4$.

3. Solve $\dfrac{x-2}{x+1} \geq 0$.

4. For $f(x) = 4 - (x-1)^2$ on $[-2; 3]$, give the maximum and the x-values where $f(x) = 0$.

5. With $A(0;1)$, $B(4;3)$, $C(6;4)$, prove that the points are collinear and give an equation of the line.

**Solution:**

1. $168/252 = 2/3$, and $2^{-3} + 1/4 = 1/8 + 2/8 = 3/8$. 2. $(x-3)^2$ ; $x - 3 = \pm 2$, so $S = \{1; 5\}$. 3. Forbidden $-1$, zero 2; signs +, −, +, so $S = ]-\infty; -1[ \cup [2; +\infty[$. 4. The square is positive or zero, so $f(x) \leq 4$, equality at $x = 1$. The zeros are $x = -1, 3$, both in the domain. 5. $\overrightarrow{AB} = (4; 2)$, $\overrightarrow{AC} = (6; 3) = \frac{3}{2}\overrightarrow{AB}$. An equation is $x - 2y + 2 = 0$.

**Guidance for the teacher**

Indicative marking: 10 points, 2 per question. For question 3, reserve one point for the domain and at the endpoints. For question 5, ask for the conclusion after the calculation.